

Faculty of Engineering

Topics in Computer Science Constraint Programming for Pandemic

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UPC

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① Solving a real-world problem

② Other Methods

③ Abstracting the problem

④ Soft Constraints

Solving a real-world problem



Solving a real-world problem

Pandemic

Hospitals must organize quickly to treat ill people.

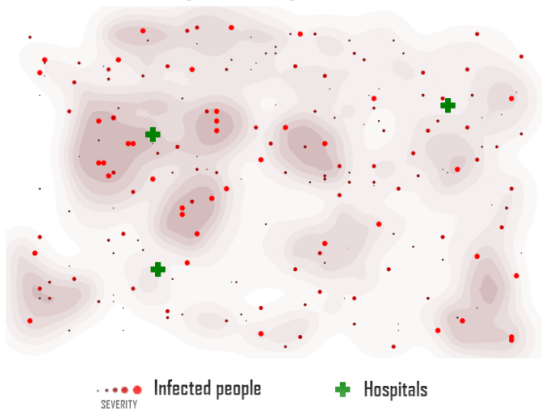


Solving a real-world problem

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Map of the pandemic



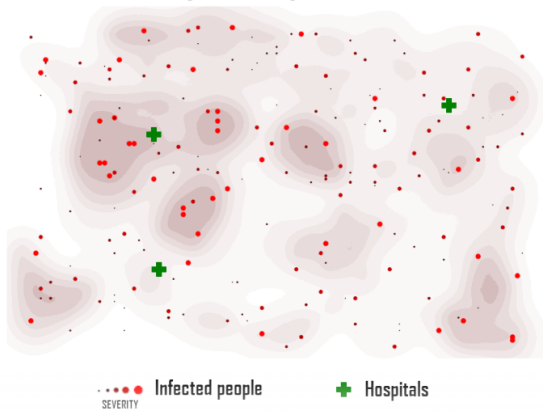


Solving a real-world problem

Pandemic

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Map of the pandemic



The world needs methods that match infected people and hospitals together given multiple criteria.

Other Methods



A broad range of methods

- SVM.
- Bayesian Networks.
- Artificial Neural Networks (ANN).

⋮



Neural Networks

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Downfalls

- The model needs to be trained, hence the need for historical data on previous cases,
- Cleaning and consolidating the dataset, takes a lot of time.
- Training and testing require lengthy sessions.



Constraint Satisfaction Problem (CSP)

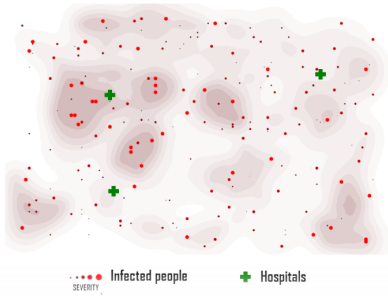
- formulating this as a CSP, won't have any of those downfalls.
- solvers can perform in nondeterministic polynomial time.
- no need for any historical data.

Abstracting the problem



Abstracting the problem

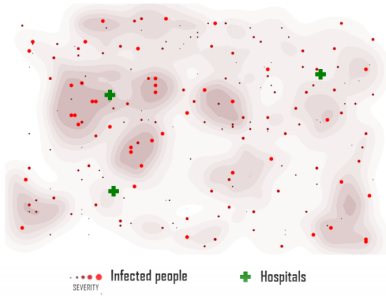
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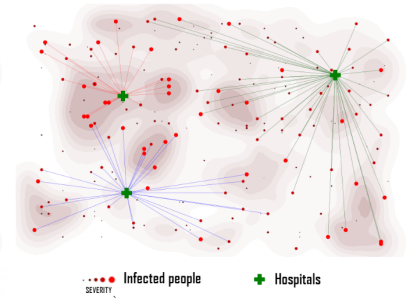


Abstracting the problem

Map of the pandemic



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Abstracting the problem

This problem can be modeled with 4 parameters:

- Location of infected people.
- Location of hospitals.
- Number of beds in each hospital.
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CSP

Variables

- H : set of hospitals.
- B_i : set of beds in hospital i .
- P : set of patients.

The set of variables is defined as:

$$x_{ijk} \in \{0, 1\}$$

where $(i, j, k) \in H \times B_i \times P$



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Constraints:

- There must be at most a single person in every bed.
 $\forall i \in H, \forall j \in B_i, \sum_{k \in P} x_{ijk} \leq 1$
- There must be at most a single bed assigned to every person.
 $\forall k \in P, \sum_{i \in H} \sum_{j \in B_i} x_{ijk} \leq 1$

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- Sick persons in severe condition should be handled first when there are not enough beds.

$$\max_{\text{sol}} \sum_{i \in B} \sum_{j \in B_i} \sum_{k \in P} x_{ijk} \times \text{sev}_k$$



CONCLUSION

A wide-angle photograph of a desert landscape. In the foreground, a vast, flat, golden-brown desert floor stretches out, marked by a single, winding, light-colored path that leads towards the horizon. A large, dark, angular rock sits on the right side of the path in the lower foreground, casting a long shadow. In the background, a range of rugged, brown mountains is visible under a clear, deep blue sky. The lighting suggests late afternoon or early morning, with long shadows and warm tones.

Beauty is born
from constraints

Leonardo Da Vinci